

Chapter 3

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet for Chapter 3: Introduction to Text-Based Programming with Python

This worksheet is designed for high school students in the course "Coding & App Development for Beginners." It covers key concepts from Chapter 3, including Python basics, environment setup, first programs, data types, variables, basic operations, and control structures.

Part 1: True/False Questions

Read each statement carefully and decide if it is True or False. Write 'T' for True or 'F' for False in the space provided. Each question is worth 2 points.

- Python is considered beginner-friendly because its syntax is complex and difficult to read. Answer: ____
- IDLE stands for Integrated Development and Learning Environment and comes with Python installation. Answer: ____
- Running the code `print("Hello, World!")` in Python will display "Hello, World!" as output. Answer: ____
- In Python, a variable can only store numbers and not text or other data types. Answer: ____
- The operator `//` in Python performs division and returns a floating-point result. Answer: ____
- An if-else statement in Python allows the program to make decisions based on conditions. Answer: ____

Part 2: Short Answer Questions

Answer the following questions in 1-2 sentences. Each question is worth 3 points.

- **Why is Python a good choice for high school students learning to code?**
- **Name one advantage of using an online editor like Replit over installing Python with IDLE on a computer.**
- **Explain what a variable is in Python and give one example.**

Part 3: Coding Exercises

Write and test the following programs in your Python environment (IDLE or online editor like Replit). Write the code in the space provided and note the output. If you can't run the code, predict the output. Each exercise is worth 5 points.

- **Personal Greeting Program** Write a program that prints a personalized greeting using your name in two separate `print()` statements. **Code:**

Write your code here

Output (or Predicted Output):

- **Simple Calculator with Variables** Create a program to calculate the total cost of buying 4 movie tickets if each ticket costs \$8.50. Store the values in variables and print the total cost. **Code:**

Write your code here

Output (or Predicted Output):

- **Decision-Making with If-Else** Write a program to check if a temperature (set to 75 degrees) is warm enough to go outside. If the temperature is above 70, print "It's warm, let's go outside!"; otherwise, print "It's too cold, stay in!" **Code:**

Write your code here

Output (or Predicted Output):

- **Looping with For Loop** Write a program using a for loop to print the numbers from 1 to 7. **Code:**

Write your code here

Output (or Predicted Output):

Part 4: Bonus Challenge (Optional)

This challenge is worth 5 extra points. Attempt it if you have time and want to push yourself further.

Mini Quiz Game Create a simple quiz program that asks the user "What is 5 + 3?" using input(). If they answer 8, print "Correct!"; otherwise, print "Wrong, the answer is 8." **Code:**

Write your code here

Output (or Predicted Output for a correct and wrong answer):

Total Points Possible: 42 (37 from required sections + 5 bonus) **Scoring Rubric:**

- True/False: 2 points each (12 total)
- Short Answers: 3 points each (9 total)
- Coding Exercises: 5 points each (20 total)
- Bonus Challenge: 5 extra points

Instructions for Submission:

- Complete all sections of this worksheet.
- If coding exercises are completed digitally, attach a screenshot or copy of the output. If done on paper, predict the output or write “Tested on [environment]” with results.

Note for Students: This worksheet is a chance to review and apply what you’ve learned in Chapter 3. Don’t worry if you make mistakes—coding is about learning through trial and error! If you’re stuck, refer to the chapter content or ask a classmate or teacher for help. Keep coding!